

INNOVATIVE STORMWATER RESILIENCY AND FINANCING MODELS

Breakout Session Summary

Westchester Clean Energy Action Conference



This breakout session paired two municipal case studies, the Village of Pelham and the City of Hoboken, to examine how local governments actually finance and build stormwater resilience. Structured half as a panel and half as a roundtable, the conversation ranged from failing infrastructure and flood mitigation engineering to municipal bonding, blended finance, public-private partnerships, and the policy tools that let communities pay for resilience without overburdening taxpayers.

SESSION PARTICIPANTS

Moderator



Tim Foley CEO/EVP
Building Reality Institute



Chance Mullen Mayor
Village of Pelham



Ravi Bhalla Managing Director | Dundon
Advisers | Former Mayor | City of Hoboken

1. NECESSITY IS THE MOTHER OF INVENTION

Mayor Mullen reframed “innovation” as something crisis forces rather than something municipalities choose. Pelham, 0.8 square miles, an \$18 million budget, entered its transformation facing a brutal inventory: a 100-year-old firehouse on rotting supports, a non-compliant police station, an inaccessible village hall, parking decks dropping concrete, sewer inflow and infiltration, and, by 2021, the highest fiscal stress score of any village in Westchester. That pressure is exactly what made bold moves possible. The throughline of the session was interconnection: expanding the tax base, fixing facilities, and mitigating floods are not separate projects but one strategy, where solving several problems at once is what reads, after the fact, as innovation.

“Necessity is the mother of invention. Innovation and transformation are ultimately made possible when you’re facing a crisis — you’re trying to solve seven problems, and then you solve two of them at the same time, and everyone goes, ‘That’s so innovative.’”

— **Chance Mullen, Mayor of Pelham**

2. BONDING IS FOUNDATIONAL — THEN LAYER EVERYTHING ELSE

Both panelists agreed that municipal bonding is the bedrock of infrastructure finance, and more important now that federal climate dollars are receding. The art is in what gets layered on top. Hoboken’s signature mechanism is a voter-approved Open Space Trust Fund, fractions of a penny per \$100 of assessed value, carved out of (not added to) the existing tax levy, that the state statute explicitly allows to be spent on debt service. That lets the city bond for parks while funding the interest without a net tax increase. On top of bonding and the trust fund, both cities stack county and state grants, FEMA programs, and private capital.

“The foundational financial mechanism to finance infrastructure is municipal bonding. That is so much more important now that the federal government is pulling out of investment in climate finance.”

— **Ravi Bhalla, Dundon Advisers; former Mayor of Hoboken**

3. CO-BENEFITS AND SOCIAL INFRASTRUCTURE

The most replicable idea was designing resilience as multi-purpose “social infrastructure.” Hoboken’s resiliency parks combine green infrastructure (rain gardens, permeable pavers, bioswales) with underground detention tanks holding up to 200,000 gallons, then discharge treated water to the Hudson, while serving day-to-day as ball fields, an ice rink, a great lawn, a cafe, and movie space. Pelham’s flood project follows the same logic: a detention system beneath a neighborhood playground, fed by larger drains and new outfalls, with a pump station and forcemain to move water uphill before it returns to the river. In both cases, the engineering solves flooding and delivers a community amenity at the same time.

4. THE RESILIENCE DIVIDEND AND CREDIT RATINGS

A key financial insight was the “resilience dividend”, and the “resiliency trap” of how differently the public and private sectors value it. The public sector sees quality-of-life returns (no wading through raw sewage, protected hospitals); the private sector sees reduced business disruption, higher property values, and lower insurance costs. Crucially, credit-rating agencies now price it in: Hoboken and Pelham both hold AA+ ratings, and Moody’s explicitly cited Hoboken’s climate-adaptation investments as a reason for confidence. Higher ratings mean cheaper borrowing, so prioritizing climate finance literally lowers the cost of doing all the other work.

5. NEGOTIATING PUBLIC-PRIVATE PARTNERSHIPS

Both cities turned private development into public infrastructure. Pelham’s “Pelham Green” deal sold an end-of-life firehouse and parking deck to a developer who, instead of paying cash, built the village a new municipal center with public parking inside a 127-unit residential building — resolving \$20.3 million of facilities needs. Hoboken used PILOT (payment-in-lieu-of-taxes) agreements to trade added density for community benefits, including a resiliency park, gym, pedestrian plaza, and decades of affordable housing, one park delivered at essentially zero taxpayer cost. The shared lesson: condition the certificate of occupancy on delivery, and bring your own financial advisor to analyze the developer’s pro forma and internal rate of return before negotiating.

6. EXPANDING THE TOOLKIT — AND OPERATING IN “NOT NORMAL TIMES”

Beyond bonding, the panel surveyed mechanisms many communities underuse: fair-share mitigation fees (assessing developers for cumulative infrastructure impact), stormwater utilities that charge by a parcel’s impervious surface rather than via general taxation, land value capture and tax-increment financing, and eminent domain to secure easements. The backdrop is federal uncertainty: FEMA would ordinarily cover half of Pelham’s ~\$48.5 million project, but “these aren’t normal times,” so the village is preparing a \$25 million bond against a hoped-for 50% county cost-share under Westchester’s Flood Mitigation Law. Hoboken’s Rebuild by Design experience showed projects can survive multiple administrations with careful, non-partisan messaging. The moderator’s closing challenge tied it back to clean energy: structure public-private partnerships that combine tax-exempt financing, solar-plus-storage, and stormwater infrastructure (e.g., solar carports over impervious lots) to create “resiliency power centers” that address flooding and grid outages together.

“There is no new housing that can be built without adequate infrastructure. Stormwater mitigation is a huge, important part of that.”

— **Tim Foley, Building and Realty Institute**

KEY TAKEAWAYS

1	Crisis creates the mandate. Pelham's worst-in-county fiscal stress and failing facilities are exactly what made transformation possible.
2	Municipal bonding is the foundation of infrastructure finance — more essential now that federal climate dollars are receding. Everything else layers on top.
3	Dedicated, voter-approved funds (e.g., Hoboken's Open Space Trust Fund) can cover bond debt service with fractions of a penny per \$100 of value — no net tax increase.
4	Design for co-benefits. Resiliency parks that store hundreds of thousands of gallons while serving as amenities solve multiple problems at once.
5	The resilience dividend is financial: climate adaptation lifted both cities to AA+ credit, lowering borrowing costs. Moody's now weighs it.
6	Public-private partnerships can deliver public assets at little or no taxpayer cost, but municipalities need their own financial advisor at the table.
7	Expand the toolkit: fair-share fees, stormwater utilities priced on impervious surface, land value capture/TIF, and carefully messaged federal programs, and pair stormwater with solar-plus-storage for resiliency power centers.